Efficient and Accurate Causal Validation of Training Data Attribution

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MATS Research

joint work with Alireza Mousavi-Hosseini and Roger Grosse

October 21, 2026

Which training data mattered, and how can we check?

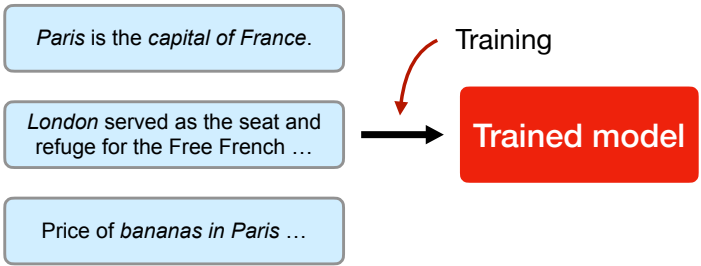
Training Data Attribution (TDA)

Paris is the *capital of France*.

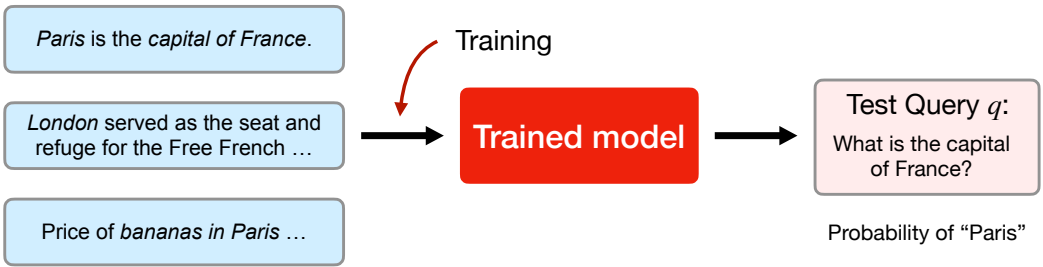
London served as the seat and refuge for the Free French …

Price of *bananas in Paris* …

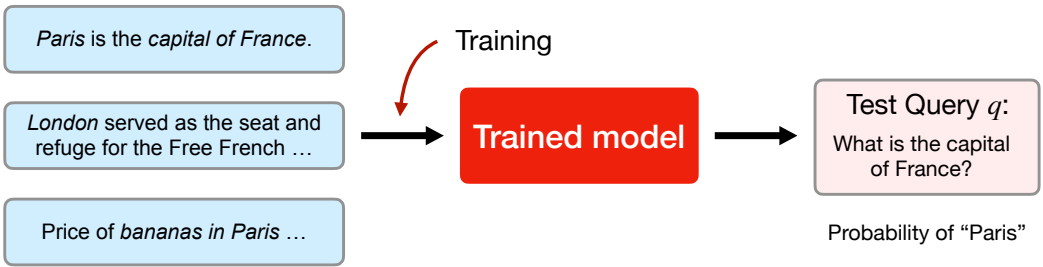
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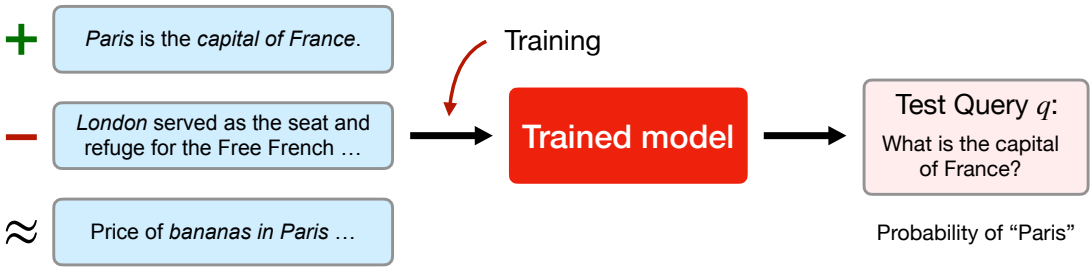


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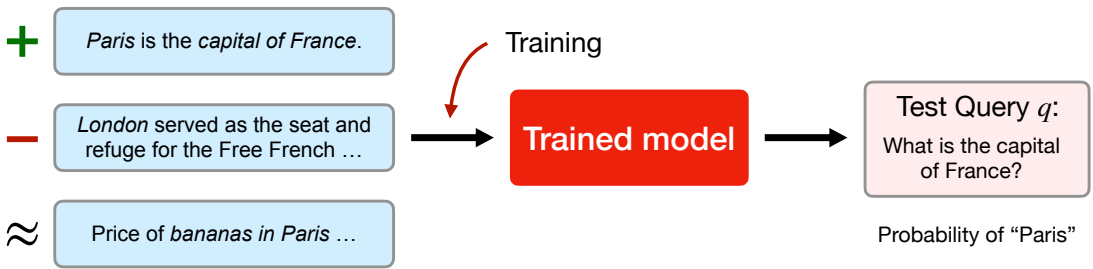
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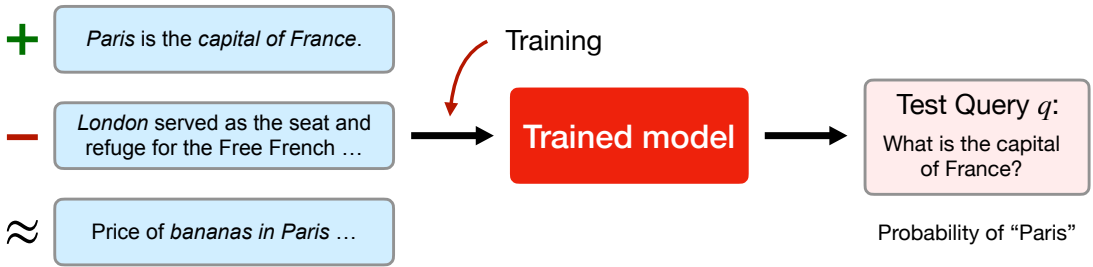
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Which training examples made “Paris” more likely?

Say $F_q\left(D \backslash \boxed{\textit{Paris} \text{ is the } \textit{capital of France.}}\right)$ is the probability of “Paris” after training without *Paris* is the *capital of France.*

Training Data Attribution (TDA)



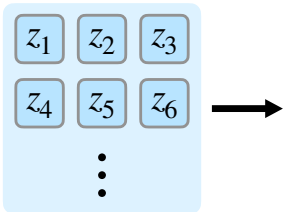
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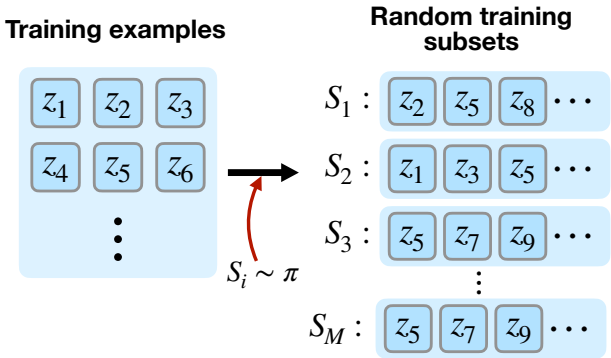
Attribution G_q estimates $F_q\left(D \setminus \boxed{\text{Paris is the capital of France.}}\right) \approx G_q\left(D \setminus \boxed{\text{Paris is the capital of France.}}\right)$.

Checking the validity of attribution

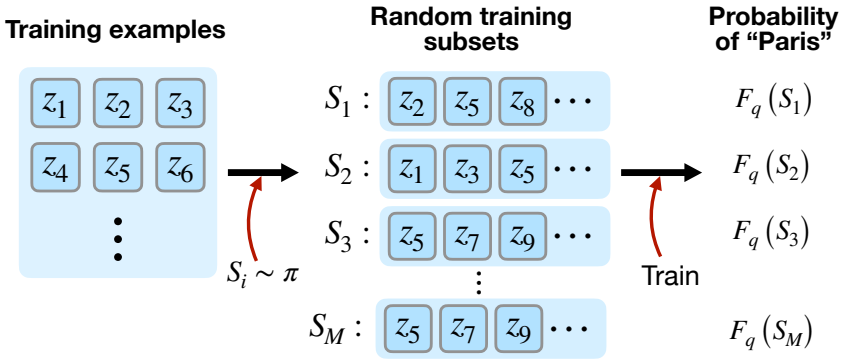
Training examples



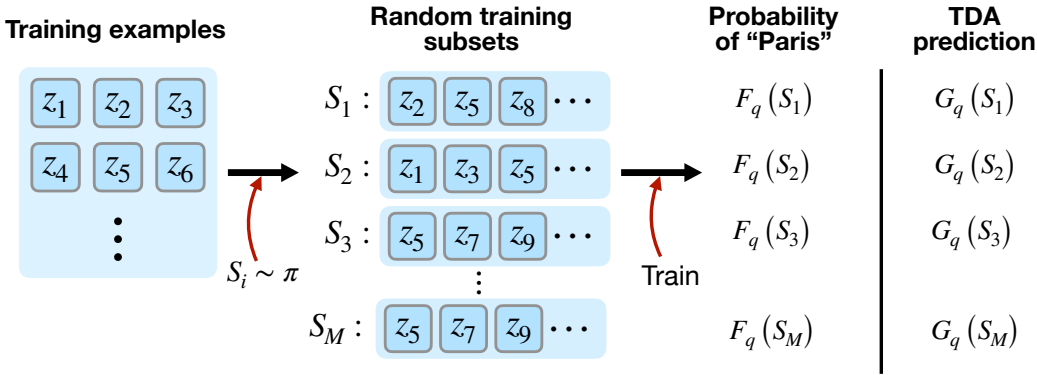
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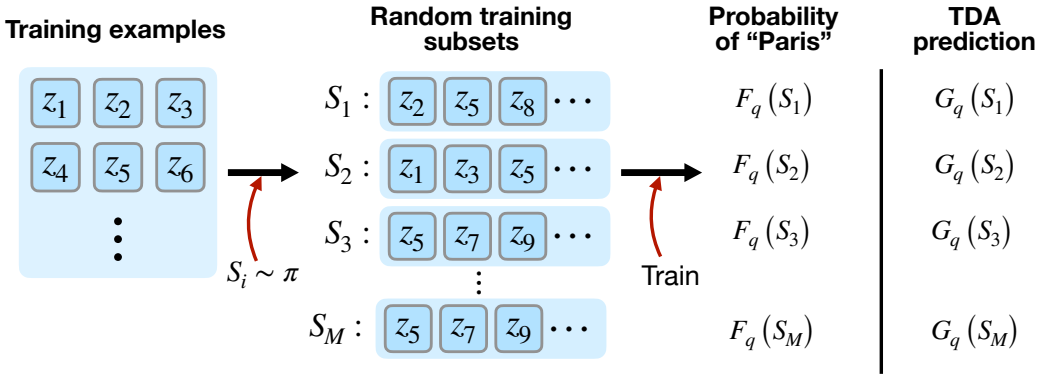
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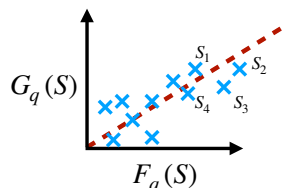


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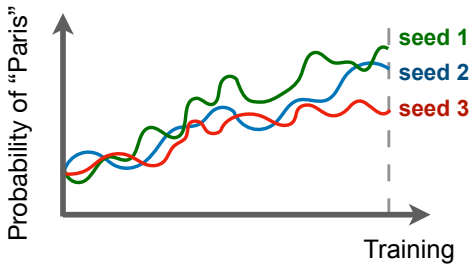
Linear Datamodeling Score

$$\text{LDS} = \frac{1}{Q} \sum_{q} \text{Corr}_{S \sim \pi} \Big(G_q(S), F_q(S) \Big)$$



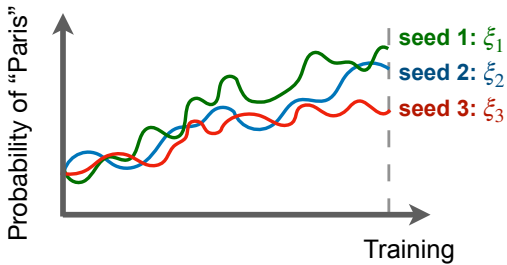
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Even for the same training set S:



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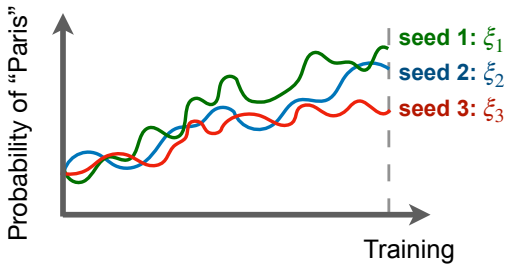


Instead of $F(S)$ we observe

$$f\left(S,\xi\right) = F(S) + \varepsilon\left(S,\xi\right)$$

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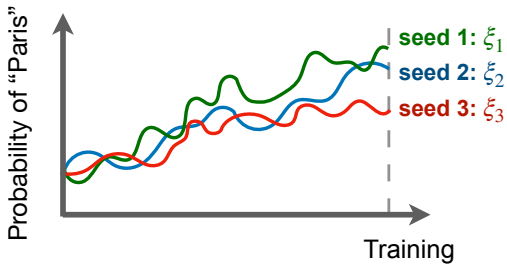
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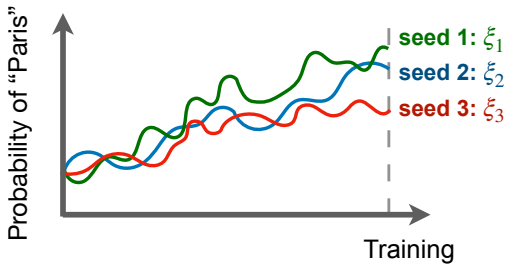
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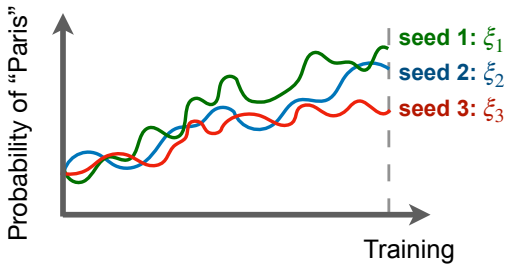
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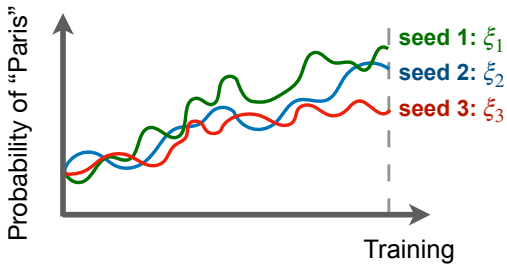
Expected data effect — **Training noise** — **Source of randomness** (data ordering, initialisation, ...)

Standard methodology: Average out R-seeds!

$$F^R(S) = \frac{1}{R}\sum_{i=1}^{R} f\left(S,\xi_i\right)$$

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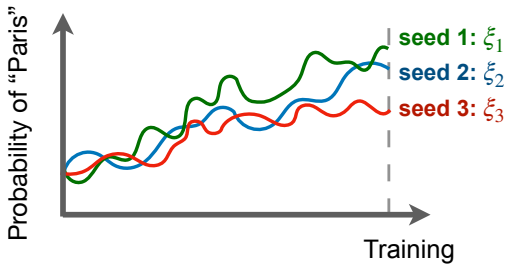
$+$

M i.i.d. subset samples

$$S_1, S_2, \ldots, S_M \sim \pi$$

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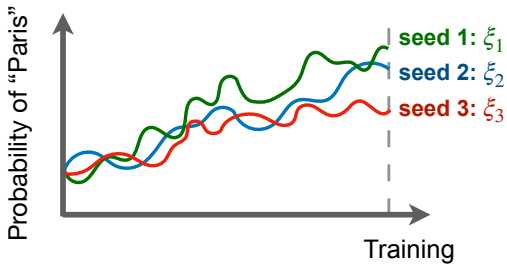
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$$S_1, S_2, \ldots, S_M \sim \pi$$

$$\Rightarrow \quad \widehat{\mathrm{LDS}}_{R,M} = \mathrm{Corr}_{S\sim\hat{\pi}_M}\left(G(S), F^R(S)\right)$$

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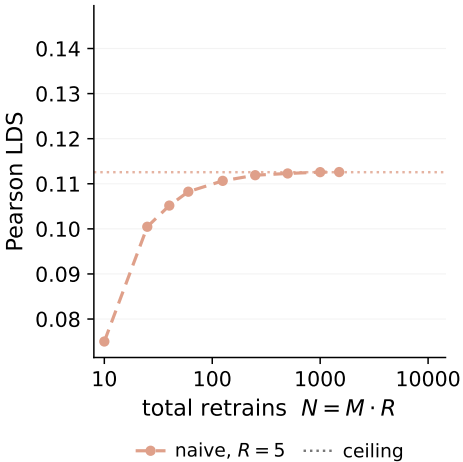
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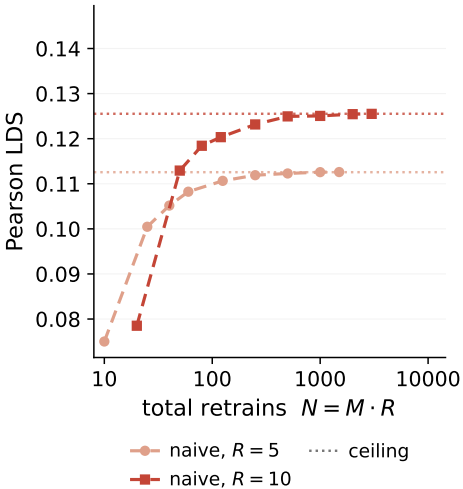
$$\widehat{\mathrm{LDS}}_{R,M} = \mathrm{Corr}_{S\sim\hat{\pi}_M}\left(G(S), F^R(S)\right)$$

Practice: $M = 100$ **and** $R = 5,10,20$

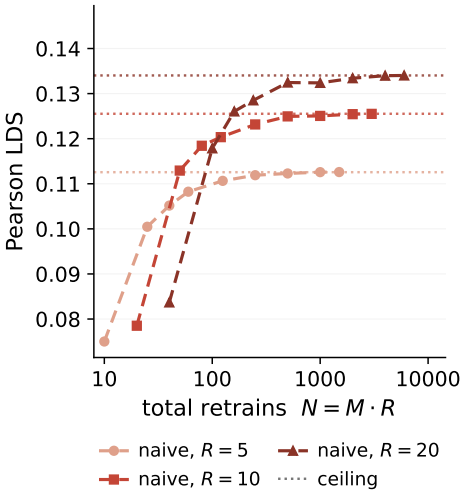
Training noise puts a ceiling on the measured score



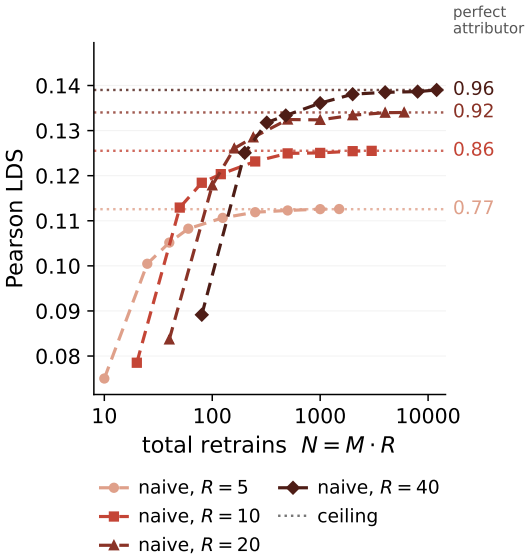
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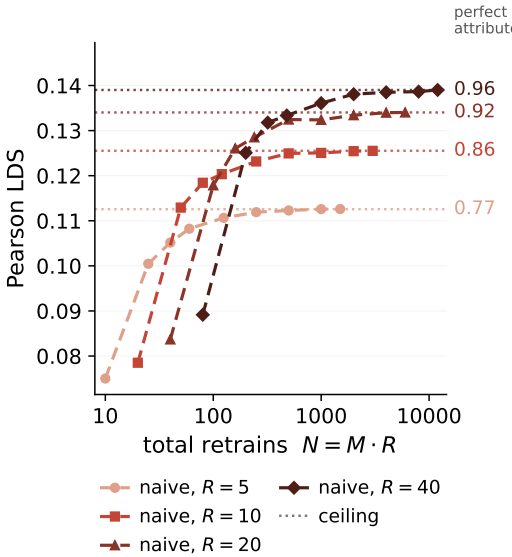
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Variance after averaging R seeds

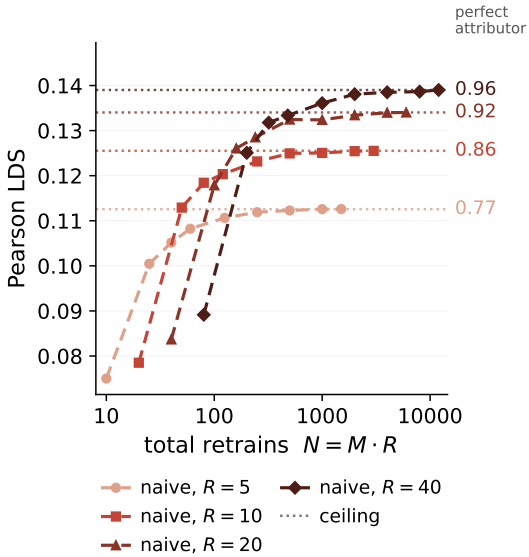
$$\mathrm{Var}_S(F^{(R)}) = \underbrace{\sigma_F^2}_{\text{data signal}} + \underbrace{\frac{\sigma_\varepsilon^2}{R}}_{\text{training noise}}$$

$\lambda = \sigma_\varepsilon^2/\sigma_F^2$: noise-to-signal ratio.

Pearson LDS ceiling

$$\mathrm{Corr}_S(G, F^{(R)}) = \frac{\mathrm{Cov}_S(G, F)}{\sqrt{\mathrm{Var}_S(G)\ \mathrm{Var}_S(F^{(R)})}}$$

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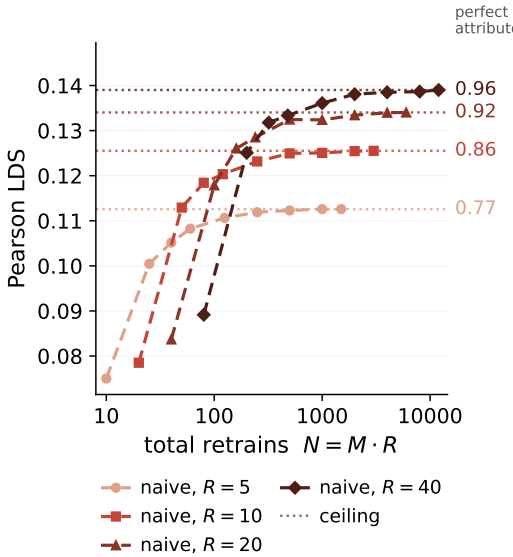
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$$\mathrm{Corr}_S(G, F^{(R)}) = \frac{\mathrm{Cov}_S(G, F)}{\sqrt{\mathrm{Var}_S(G)\left(\sigma_F^2 + \sigma_\varepsilon^2/R\right)}}$$

Training noise increases the denominator.

Training noise puts a ceiling on the measured score



Variance after averaging R seeds

$$\mathrm{Var}_S(F^{(R)}) = \underbrace{\sigma_F^2}_{\text{data signal}} + \underbrace{\frac{\sigma_\varepsilon^2}{R}}_{\text{training noise}}$$

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Pearson LDS ceiling

$$\mathrm{Corr}_S(G, F^{(R)}) = \frac{\mathrm{Corr}_S(G, F)}{\sqrt{1 + \lambda/R}}.$$

$\mathrm{Corr}_S(G, F)$: score without training noise.

More subsets M do not lift this ceiling.

Repeated runs let us correct the score

Same subset, different seeds

$$s^2(S) = \widehat{\mathrm{Var}}_{\xi}\big[f(S, \xi)\big]$$

Sample variance over R runs

Average across subsets

$$\widehat{\sigma}^2_{\varepsilon} = \frac{s^2(S_1) + \cdots + s^2(S_M)}{M}$$

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Subtract the noise

$$\widehat{\sigma}^2_F = \widehat{\mathrm{Var}}_S(F^{(R)}) - \frac{\widehat{\sigma}^2_\varepsilon}{R}.$$

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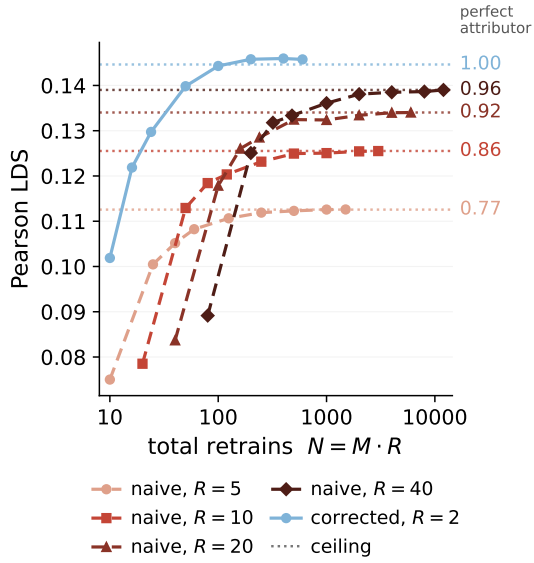
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Just $R = 2$ runs per subset.



Estimating the biased LDS with fewer retrainings

Estimate from two runs

$$\underbrace{\widehat{\sigma}_F^2}_{\text{data signal}} \qquad \underbrace{\widehat{\sigma}_\varepsilon^2}_{\text{training noise}}$$

Reconstruct five-seed variance

$$\widehat{\mathrm{Var}}_S(F^{(5)}) = \widehat{\sigma}_F^2 + \frac{\widehat{\sigma}_\varepsilon^2}{5}$$

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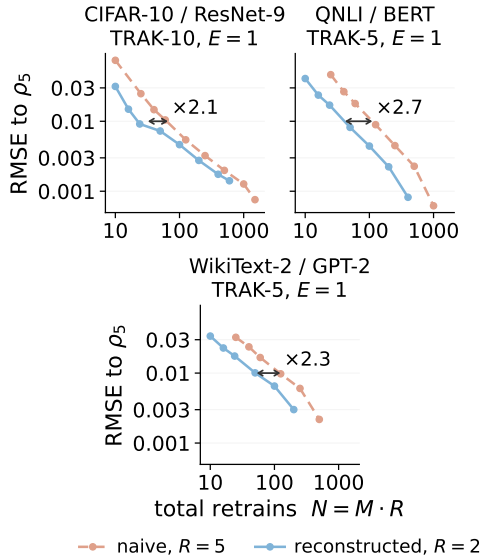
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2–5× fewer retrainings

across CIFAR-10, QNLI and WikiText-2



Forecasting larger attribution ensembles

$$g(S,\zeta) = \underbrace{G(S)}_{\text{mean}} + \underbrace{\gamma(S,\zeta)}_{\text{zero-mean noise}}$$

$$G^{(E)}(S) = \frac{1}{E}\sum_{i=1}^{E} g(S,\zeta_i)$$

E independent members.

$$\widehat{c} = \underbrace{\widehat{\sigma}^2_\gamma}_{\text{attribution noise}} \Big/ \underbrace{\widehat{\sigma}^2_G}_{\text{data signal}}$$

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Forecast from two members

$$\widehat{\rho}_E^{\text{forec}} = \widehat{\text{Corr}}(G^{(2)}, F)\sqrt{\frac{1+\widehat{c}/2}{1+\widehat{c}/E}}$$

$\widehat{\text{Corr}}$: corrected Pearson LDS.

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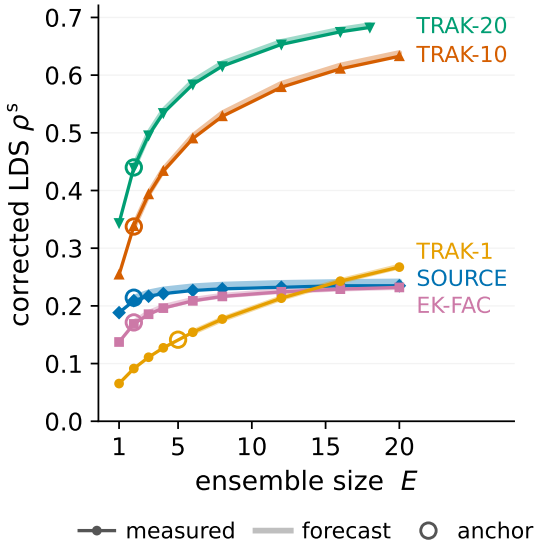
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Conclusions

Main takeaways

- Training noise can hide attribution signal.
- 2–5× **fewer retrainings** at matched error on the five-seed benchmark.
- Small ensembles can forecast larger ensembles' scores.

Next steps / questions

- How much of this translates to downstream tasks?
- Other metrics reflecting the magnitude of an intervention?

Code: github.com/simonvary/corrected-lds

Data: huggingface.co/datasets/corrected-lds/data